



# **DATA MODELLING**

# Introduction to Data Models

I want to Build a  
Hybrid  
Automobile

I want to build  
a mansion

I want to build  
a Spaceship

I want to build an  
Operating  
System

I want to build  
an Application

I want to build  
a Database

I want to build  
a Warehouse



What do I need  
to Accomplish my Goal?

You Need a **Model**

# Definition of a Model

- **Model** is a replica or a representation of particular aspects and segments of the real world.
- **Modeling** provides effective ways to describe/verify the real-world information requirements to/from the stakeholders in an organization.
- Modeling is an **integral part of the design and development of any system.**
- A **correct model** is essential.

# DATA MODELLING - Cont..,

- Generally speaking, a model is an abstraction and reflection of the real world.
- Modeling gives us the ability to visualize what we cannot yet realize.  
It is the same with data modeling.
- The primary aim of a data model is to make sure that all data objects required by the business are accurately and fully represented.

# Data Modeling

## ➤ WHAT IS A DATA MODEL?

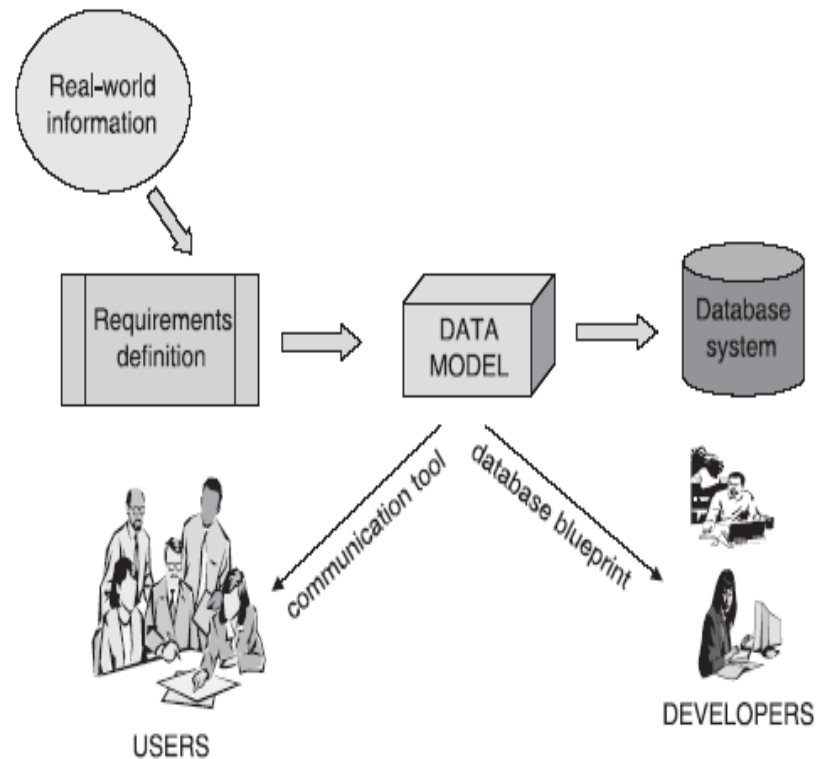
A data model is an abstraction of some aspect of the real world (system).

## ➤ NEED OF DATA MODEL?

- Helps to visualize the business
- A model is a means of communication.
- Models help elicit and document requirements.
- Models reduce the cost of change.
- Model is the essence of DW architecture based on which DW will be implemented

# What is Data Modeling?

- Data modeling is a technique for exploring the data structures needed to support an organization's information need.
- It would be a conceptual representation or a replica of the data structure required in the database system.
- A data model focuses on which data is required and how the data should be organized.
- At the conceptual level, the data model is independent of any hardware or software constraints.



# DATA MODELLING – Cont..,

- Data modeling is important because it specifies the data structure, which can impact all aspects of data usage.
- For example, it can have a significant impact on performance. This is particularly true with data warehousing.
- And, the data warehouse is the primary structural element in business intelligence.

# What do we want to do with the data?

Model depends on what kind of data analysis we want to do:

- **Different Data Analysis Techniques**
  - ✓ Query and reporting
    - Display Query Results
  - ✓ Multidimensional analysis
    - Analyze data content by looking at it in different perspectives
  - ✓ Data mining
    - Discover patterns and clustering attributes in data

# Example of a Simple Data Model

- The data is divided into two tables: one for policy data and one for customer data.

POLICY TABLE

| Policy Number | Date Issued | Policy Type | Customer Number | Commission Rate | Maturity Date |
|---------------|-------------|-------------|-----------------|-----------------|---------------|
| V213748       | 02/29/1989  | E20         | HAYES01         | 12%             | 02/29/2009    |
| N065987       | 04/04/1984  | E20         | WALSH01         | 12%             | 04/04/2004    |
| W345798       | 12/18/1987  | WOL         | ODEAJ13         | 8%              | 06/12/2047    |
| W678649       | 09/12/1967  | WOL         | RICHB76         | 8%              | 09/12/2006    |
| V986377       | 11/07/1977  | SUI         | RICHB76         | 14%             | 09/12/2006    |

CUSTOMER TABLE

| Customer Number | Name    | Address         | Postal Code | Gender | Age | Birth Date |
|-----------------|---------|-----------------|-------------|--------|-----|------------|
| HAYES01         | S Hayes | 3/1 Collins St  | 3000        | F      | 25  | 06/23/1975 |
| WALSH01         | H Walsh | 2 Allen Road    | 3065        | M      | 53  | 04/16/1947 |
| ODEAJ13         | J O'Dea | 69 Black Street | 3145        | M      | 33  | 06/12/1967 |
| RICHB76         | B Rich  | 181 Kemp Rd     | 3507        | M      | 59  | 09/12/1941 |

# What Makes a Good Data Model?

- **Completeness**

- Ensure that every piece of information required for a System is recorded and maintained.

- **Non-Redundant**

- One fact should be recorded only once. Repetition may result in inconsistency and increased storage requirements.

# What Makes a Good Data Model? (contd..)

- Adherence to Business Rules
  - Ensure that every piece of information required for a System is recorded and maintained.
  - The collected data is to be recorded by considering all business rules. It should not violate any rule.

# What Makes a Good Data Model? (contd..)

- Data Reusability
  - Design a data structure to ensure re-usability.
- Stability and Flexibility
  - A model needs to be flexible enough to adopt to new changes without forcing the programmer to re-write the code.

# What Makes a Good Data Model? (contd..)

- Elegance

- A data model should neatly present the required data in the least possible number of groups or tables.

- Communication

- A model should present the data in a manner understandable to all stakeholders.

# What Makes a Good Data Model? (contd..)

- Integration

- A good model is compatible with the existing and future systems.

- Avoid Conflicting Objectives

- A good model can strike a good balance between groups with different sets of requirements.

# Performance of a Data Model

- Performance makes a good model better...
- Performance differs from our other criteria because it depends heavily on the software and hardware platforms on which the database will run.
- Performance requirements are usually “added to the mix” at a stage later than the other criteria, only when necessary.

# Where Data Models are used ?

- Operational Systems
  - Traditional Applications designed to run the day-to-day business of the Enterprise
- External Systems \*\*\*
  - Data used within an Enterprise that is obtained from outside sources
- Staging Areas \*\*\*
  - Created to aid in the collection and transformation of data that is targeted for a Data Warehouse

# Where Data Models are used ?

- Operational Data Store \*\*\*

- W. H. Inmon and Claudia Imhoff definition: “A subject-oriented, integrated, volatile, current valued data store containing only corporate detailed data”.

- Data Warehouse (DW)

- W. H. Inmon definition: “A subject-oriented, integrated, non-volatile, time-variant collection of data organized to support management needs”.

- Data Mart (DM)

- TDWI definition: “A data structure that is optimized for access. It is designed to facilitate end-user analysis of data. It typically supports a single analytic application used by a distinct set of workers.”

- \*\*\* - Not discussed here

# What Data Modeling is not...

- A waste of time!
- A one time effort
- The ultimate IT application development cure
- A quick process
- A function solely performed and understood by and for IT professionals

# People involved in Data Modeling

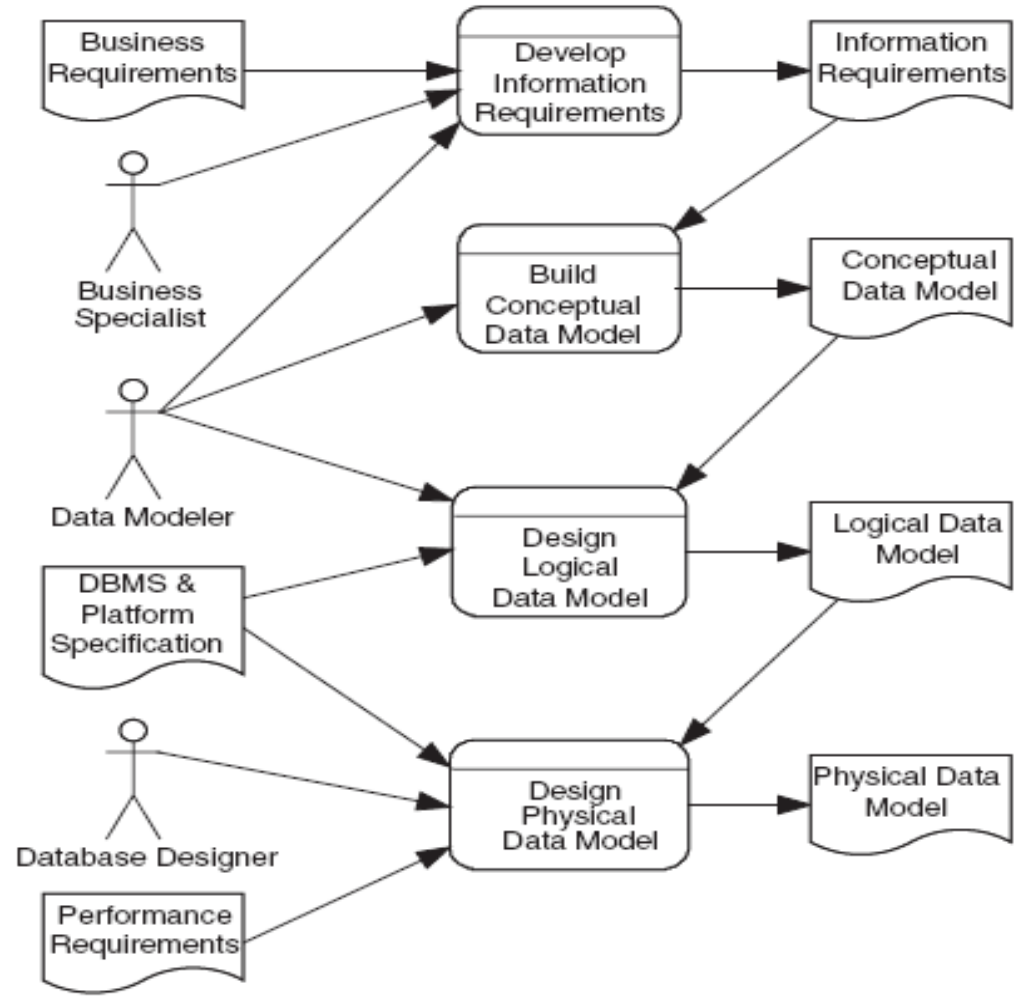
- System users, owners, and/or sponsors of business
  - To verify that the model meets their requirements..
- Business specialists (subject matter experts or SMEs)
  - To verify the accuracy and stability of the business rule and processes.
- Data modeler
  - To ensure that he will design the model correctly and will not miss out on any important requirement.
- Process modelers
  - To ensure that they will use the model correctly.

# People involved in Data Modeling (contd..)

- Physical database designer (or DBA)
  - To understand the difference between logical and physical model
  - To design database to achieve the required performance
- Systems integration manager and enterprise architect
  - To understand how the new database will fit into existing system.
  - To think beyond current project.

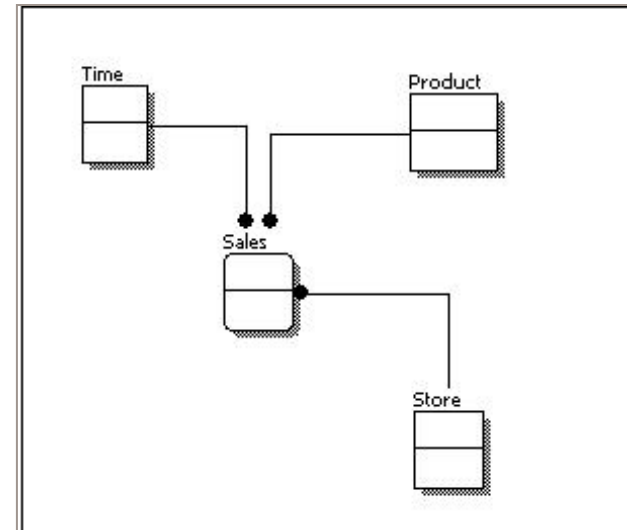
# Data modeling stages and deliverables

- A data modeling process goes through various stages and produces the following deliverables:
  - Conceptual Model
  - Logical Model
  - Physical Data Model



# Conceptual Data Model

- A conceptual data model identifies the highest-level relationships between the different entities
  - Features of conceptual data model include:
    - Includes the important entities and the relationships among them.
    - No attribute is specified.
    - No primary key is specified

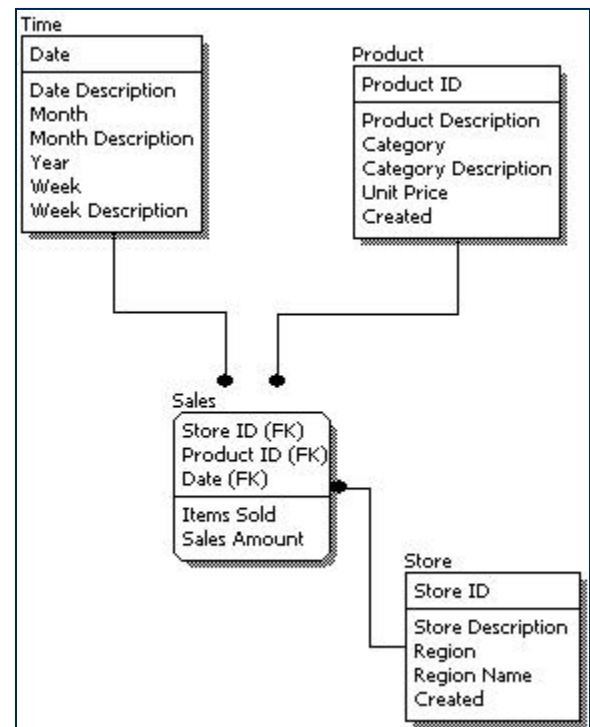


# Logical Data Model

- A logical data model describes the data in as much detail as possible, without regard to how they will be physical implemented in the database.
- Features of a logical data model include:
  - Includes all entities and relationships among them.
  - All attributes for each entity are specified.
  - The primary key for each entity is specified.
  - Foreign keys (keys identifying the relationship between different entities) are specified.
  - Normalization occurs at this level.

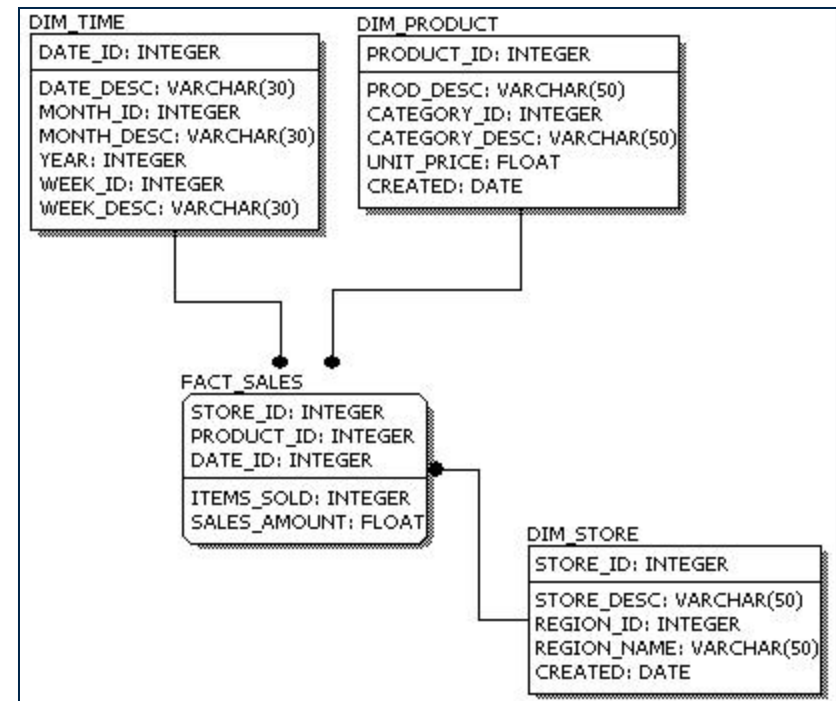
# Logical Data Model (contd..)

- The steps for designing the logical data model are as follows:
  - Specify primary keys for all entities.
  - Find the relationships between different entities.
  - Find all attributes for each entity.
  - Resolve many-to-many relationships.
  - Normalization.



# Physical Data Model (contd..)

- Physical data model represents how the model will be built in the database.
- A physical database model shows all table structures, including column name, column data type, column constraints, primary key, foreign key, and relationships between tables.



# Physical Data Model (contd..)

- Features of a physical data model include:
  - Specification of all tables and columns.
  - Foreign keys are used to identify relationships between tables.
  - Demoralization may occur based on user requirements.
  - Physical considerations may cause the physical data model to be quite different from the logical data model.
  - Physical data model will be different for different RDBMS. For example, data type for a column may be different between MySQL and SQL Server

# Levels of Information

## Conceptual

- Information Content - General Ideas
- Human Concept of Application Domain
- Data System as Understood by Users

## Logical

- Details of whole information Content
- Reference to specific Database Software
- No Details of Hardware/Software

## Physical

- Details at Level of Internal data storage
- Intricacies of Specific Database
- Details of Physical implementation

# **Data Modeling for Business Intelligence**

## **Lesson 2: Understanding Business Requirements**

# Lesson Objectives

- This lesson will provide overview of various techniques of Requirement gathering
- We will learn about:
  - Need of Requirement Analysis
  - The Data Life cycle
  - Ways of Collecting requirement
  - Business Requirement Specification (BRS)

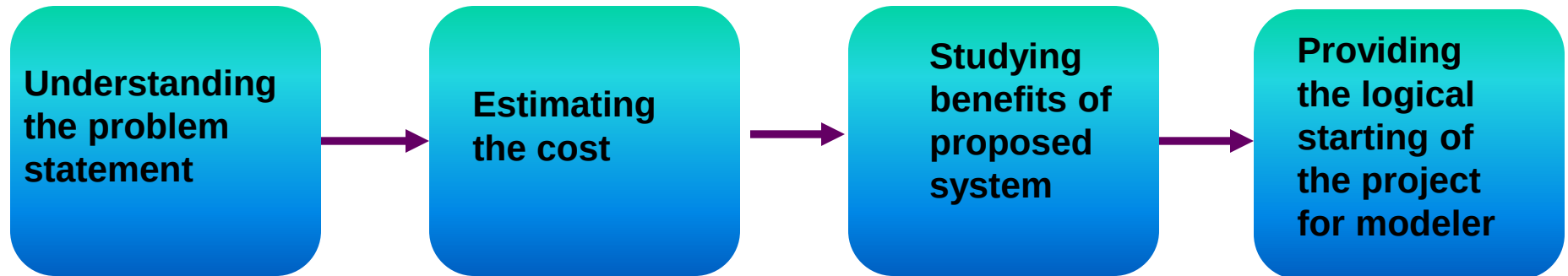


# Requirements Collection

- Experts think that the requirement gathering should be treated as a separate phase.
- Though, some suggest that it should be a part of the conceptual design phase.
- The requirement phase is used for the following:
  - Collecting the business requirement
  - Formulating the understanding of requirement
- Requirement analysis starts as soon as a business case is prepared or received.

# Understanding Business Requirements

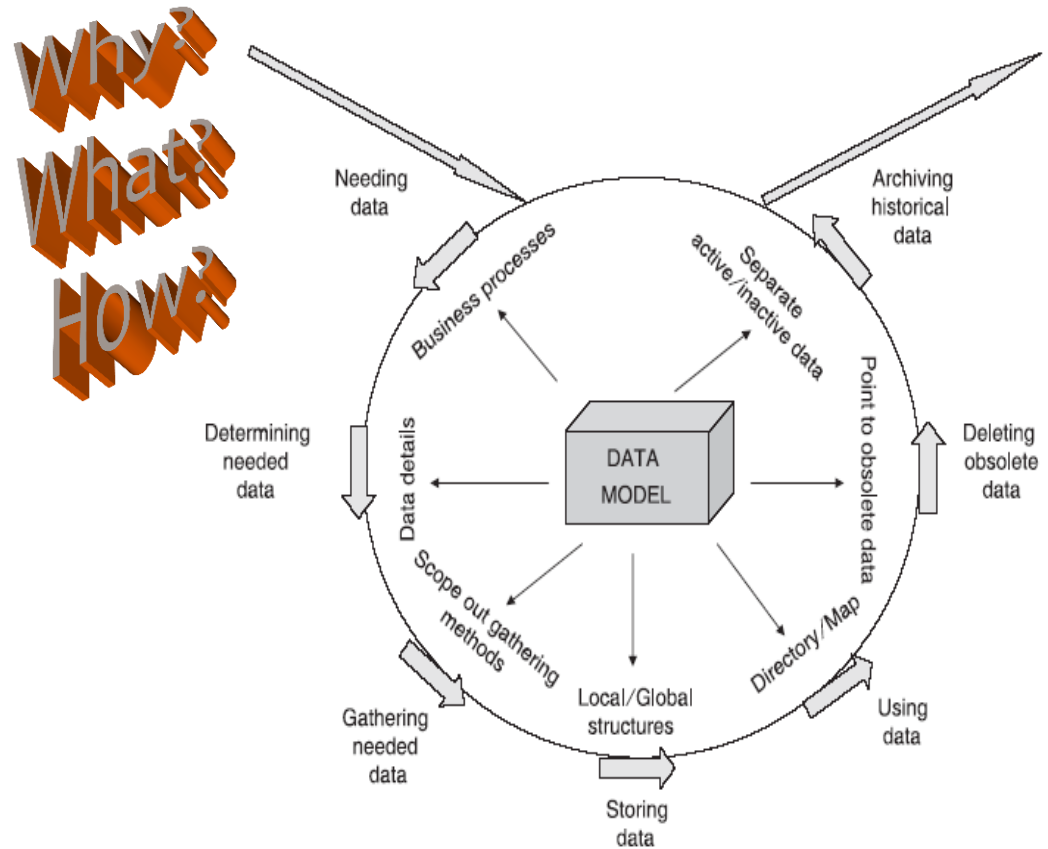
- Any system is usually developed in response to a problem, an opportunity, or a requirement.
- Its statement should be supported by a formal business case. The case is used for the following:



- It is important to understand the data life cycle in an application.

# Study of Data Life Cycle

- Need for data
  - Needed data
  - Collect needed data
  - Store data
  - Use data
  - Delete obsolete data
  - Archive historical data
- 
- A data life cycle help us to state the requirements clearly.



# Characteristics of a Good Requirement

- **Specific**
  - **Correct**: A true statement of what the requirement should do
  - **Complete**: Encompass all requirements of concern to the Users
  - **Unambiguous**: Has only one interpretation
- **Consistent**: Does not conflict with other Requirements
- **Verifiable**: Can be tested to meet the Requirements
- **Attainable**: It should be within the scope of the project
- **Understandable**:  
Comprehensible by User, Business and Developers
- **Detailed/ Granular**: Granular to be implemented in test cases and design
- **Explicit**: Encompass all derived requirements
- **Traceable**: It should be possible to trace a component requirement to its source
- **Manageable & Organized**:  
Scalability and change management, should be structured

# Collation of Business Requirements

- Conduct interviews and workshops:
  - Avoid using data model in interviews and workshops.
  - Prefer UML, Use Cases, Activity Diagrams, DFD, and so on.
  - Conduct interviews with senior managers.
  - Conduct interviews with Subject Matter Experts (Do not let them Design.)
  - Conduct facilitated workshops.



- Verify your own understanding about requirements.



# Interviewing Stakeholders and Users

- Ask questions to the stakeholders at a pre-decided time and venue to gather requirement knowledge:
  - Ask open-ended questions
  - Use structured agenda of fairly open questions
- Interviews are good for documentation and agreement on common or discussed objectives.
- Management support is required to obtain time from stakeholders.

# Other Methods of Collecting Requirements

- **Direct Observation Techniques:**

- This allows you to assess users' needs and problems associated with the use of services.
- This technique is designed for a specific purpose; to identify a problem, describe a situation, assess user satisfaction, and so on.

- **Surveys**

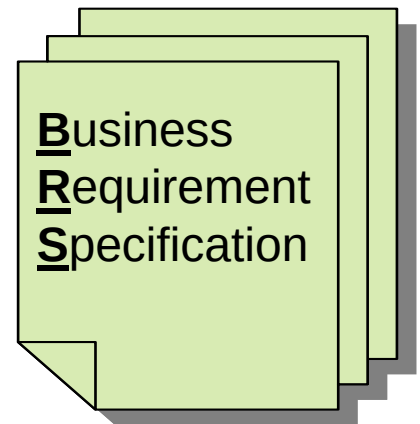
- This is more suitable when stakeholders are spread globally.

- **Data Collection and Analysis**

- These are indirect sources of information to provide an approximation of the needs of the user.
- Source can be public data, marketing data or any other data.

# Business Requirements Specification

- **The most important task is to define “statement of requirements” or Business Requirement Specification.**
- The issues could be as follows:
  - Many requirements are well-known but impractical to document them.
  - Some requirements are only relevant to specific design alternatives.
  - Some requirements may emerge only when the client has seen an actual design.
  - High-level business directions and rules cannot be captured directly.



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# **DATA MODELING TECHNIQUES**

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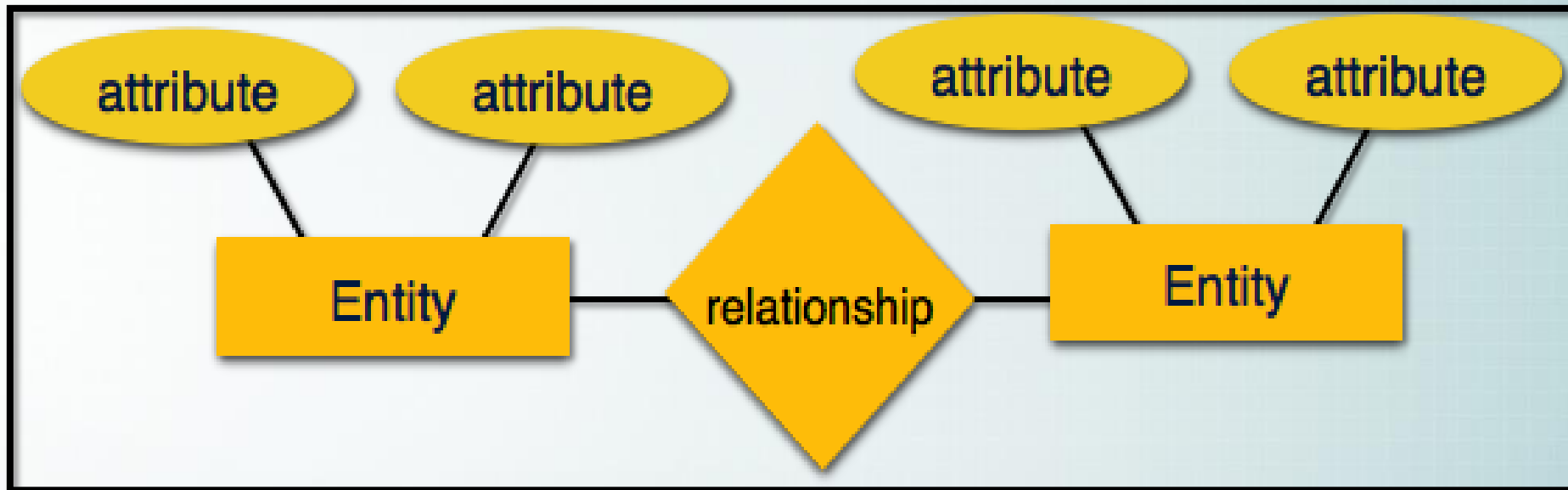
- **There are 2 techniques of Data Modeling :**
  - **E-R Modeling**
  - **Dimensional Modeling**

# Entity-Relationship Model

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- **Entity-Relationship (ER) Model is based on the notion of real-world entities and relationships among them.**
- **While formulating real-world scenario into the database model, the ER Model creates entity set, relationship set, general attributes and constraints.**
- **ER Model is best used for the conceptual design of a database. ER Model is based on the following:**
  - **Entities** and their *attributes*.
  - **Relationships** among **entities**.

# ER Model - Example Notation



# ENTITY and RELATIONSHIP

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- **Entity** – An entity in an ER Model is a real-world entity having properties called **attributes**.
- Every **attribute** is defined by its set of values called **domain**.
- **For example**, in a school database, a student is considered as an entity. Student has various attributes like name, age, class, etc.
- **Relationship** – The logical association among entities is called **relationship**.
- Relationships are mapped with entities in various ways. Mapping cardinalities define the number of association between two entities.

# TYPES OF DATAMODEL

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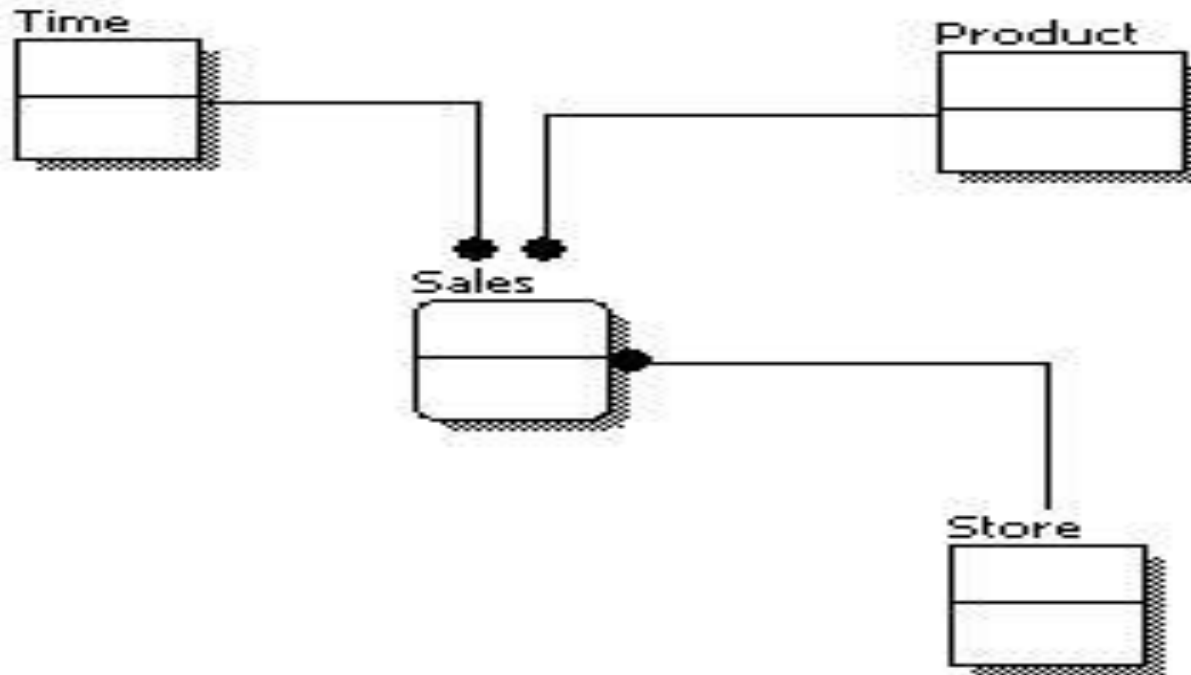
- **There are 3 types of Data Models as listed below:**
  - **CONCEPTUAL DATA MODEL**
  - **LOGICAL DATA MODEL**
  - **PHYSICAL DATA MODEL**

# CONCEPTUAL DATA MODEL

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- **A conceptual data model identifies the highest-level relationships between the different entities.**
- **Features of conceptual data model include:**
  - Includes the important entities and the relationships among them.
  - No attribute is specified.
  - No primary key is specified.

# CONCEPTUAL DATA MODEL



**From the figure above, we can see that the only information shown via the conceptual data model is the entities that describe the data and the relationships between those entities. No other information is shown through the conceptual data model.**

# LOGICAL DATA MODEL

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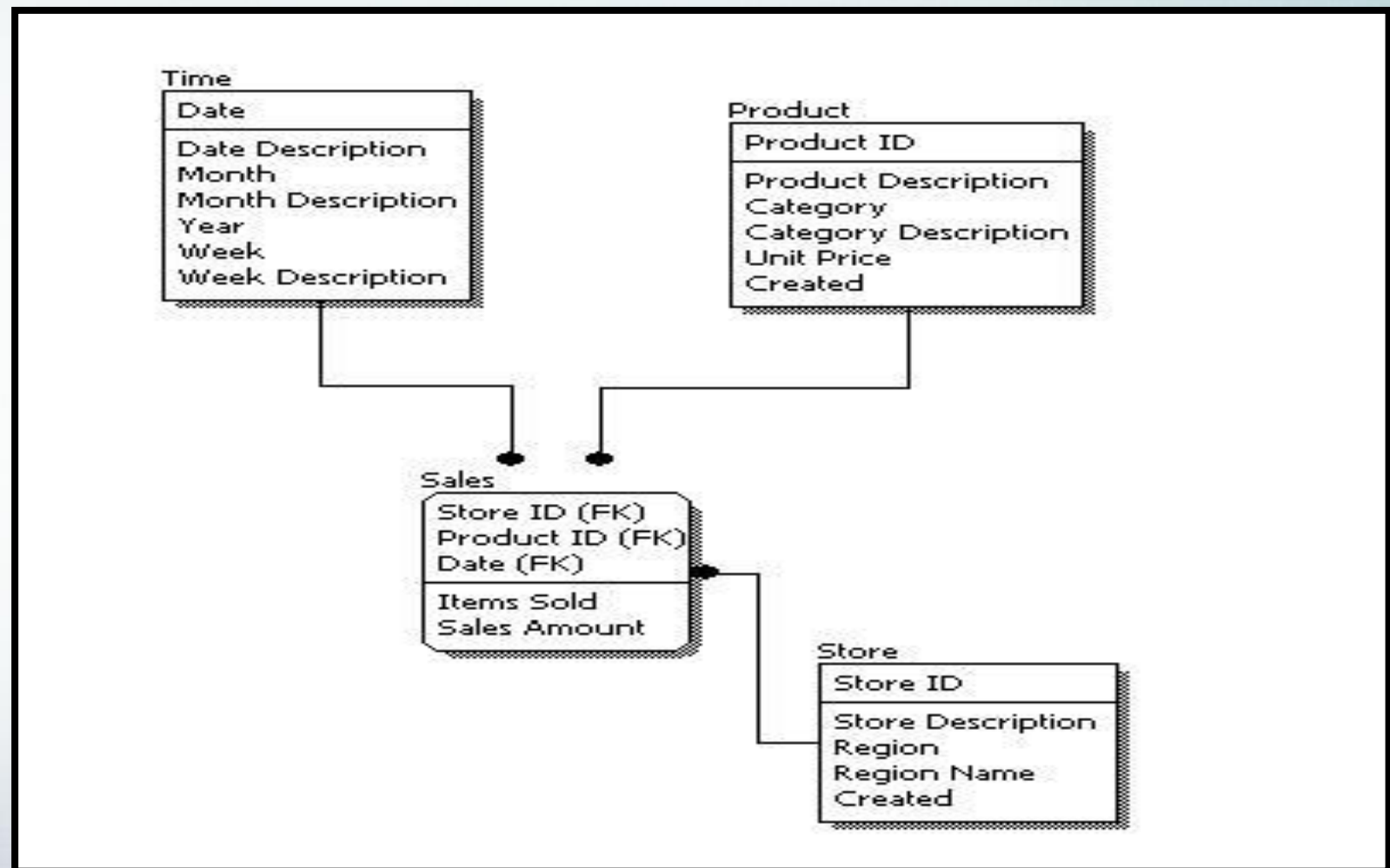
- A logical data model describes the data in as much detail as possible, without regard to how they will be physical implemented in the database.
- **Features of a logical data model include:**
  - Includes all entities and relationships among them.
  - All attributes for each entity are specified.
  - The primary key for each entity is specified.
  - Foreign keys (keys identifying the relationship between different entities) are specified.
  - Normalization occurs at this level.

# Steps for Logical Data Model Design

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- The steps for designing the logical data model are as follows:
  - *Specify primary keys for all entities.*
  - *Find the relationships between different entities.*
  - *Find all attributes for each entity.*
  - *Resolve many-to-many relationships.*
  - *Normalization.*

# LOGICAL DATA MODEL



# COMPARISON LOGICAL '&' CONCEPTUAL

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- ***Comparing the logical data model shown above with the conceptual data model diagram, we see the main differences between the two:***
- **In a logical data model, primary keys are present, whereas in a conceptual data model, no primary key is present.**
- **In a logical data model, all attributes are specified within an entity. No attributes are specified in a conceptual data model.**

# COMPARISON LOGICAL '&' CONCEPTUAL

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- **Relationships between entities are specified using primary keys and foreign keys in a logical data model.**
- **In a conceptual data model, the relationships are simply stated, not specified, so we simply know that two entities are related, but we do not specify what attributes are used for this relationship.**

# PHYSICAL DATA MODEL

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- **Physical data model represents how the model will be built in the database.**
- **A physical database model shows all table structures, including column name, column data type, column constraints, primary key, foreign key, and relationships between tables.**
- **Features of a physical data model include:**
  - **Specification all tables and columns.**
  - **Foreign keys are used to identify relationships between tables.**

# PHYSICAL DATA MODEL

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- **Denormalization may occur based on user requirements.**
- **Physical considerations may cause the physical data model to be quite different from the logical data model.**
- **Physical data model will be different for different RDBMS. For example, data type for a column may be different between MySQL and SQL Server.**

# STEPS FOR DESIGNING PHYSICAL DM

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- **The steps for physical data model design are as follows:**
  - **Convert entities into tables.**
  - **Convert relationships into foreign keys.**
  - **Convert attributes into columns.**
  - **Modify the physical data model based on physical constraints / requirements.**

# PHYSICAL DATA MODEL

## DIM\_TIME

|                         |
|-------------------------|
| DATE_ID: INTEGER        |
| DATE_DESC: VARCHAR(30)  |
| MONTH_ID: INTEGER       |
| MONTH_DESC: VARCHAR(30) |
| YEAR: INTEGER           |
| WEEK_ID: INTEGER        |
| WEEK_DESC: VARCHAR(30)  |

## DIM\_PRODUCT

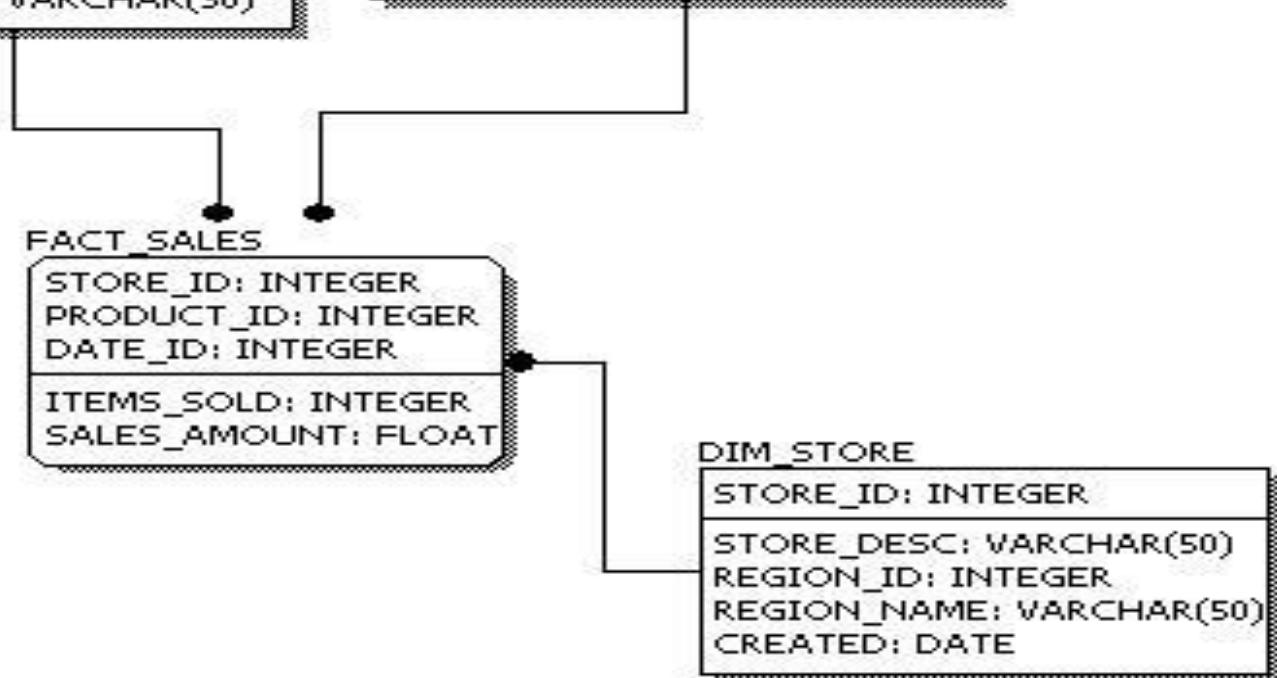
|                            |
|----------------------------|
| PRODUCT_ID: INTEGER        |
| PROD_DESC: VARCHAR(50)     |
| CATEGORY_ID: INTEGER       |
| CATEGORY_DESC: VARCHAR(50) |
| UNIT_PRICE: FLOAT          |
| CREATED: DATE              |

## FACT\_SALES

|                     |
|---------------------|
| STORE_ID: INTEGER   |
| PRODUCT_ID: INTEGER |
| DATE_ID: INTEGER    |
| ITEMS_SOLD: INTEGER |
| SALES_AMOUNT: FLOAT |

## DIM\_STORE

|                          |
|--------------------------|
| STORE_ID: INTEGER        |
| STORE_DESC: VARCHAR(50)  |
| REGION_ID: INTEGER       |
| REGION_NAME: VARCHAR(50) |
| CREATED: DATE            |



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# **CONCEPTS OF ER - MODELLING**

# CONCEPTS OF THE ER MODEL

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- **Entity types**
- **Relationship types**
- **Attributes**

# ENTITY TYPE

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## ➤ Entity type

- Group of objects with same properties, identified by enterprise as having an independent existence.
- **For example**, in a school database, students, teachers, classes, and courses offered can be considered as entities.

## ➤ Entity occurrence

- Uniquely identifiable object of an entity type.

# Example: For an ENTITY

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Student

Teacher

Projects

# ER - MODELLING SYMBOLS

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- ENTITY SET



- WEAK ENTITY SET



- RELATIONSHIP SET



- IDENTIFYING RELATIONSHIP FOR A  
WEAK ENTITY SET

# ER - MODELLING SYMBOLS

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- **ATTRIBUTE**



- **MULTI-VALUED ATTRIBUTE**



- **DERIVED ATTRIBUTE**



- **PRIMARY KEY ATTRIBUTE**

# ER - MODELLING SYMBOLS

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- ONE-TO-ONE RELATIONSHIP



- MANY-TO-ONE RELATIONSHIP



- MANY-TO-MANY RELATIONSHIP

# ATTRIBUTES

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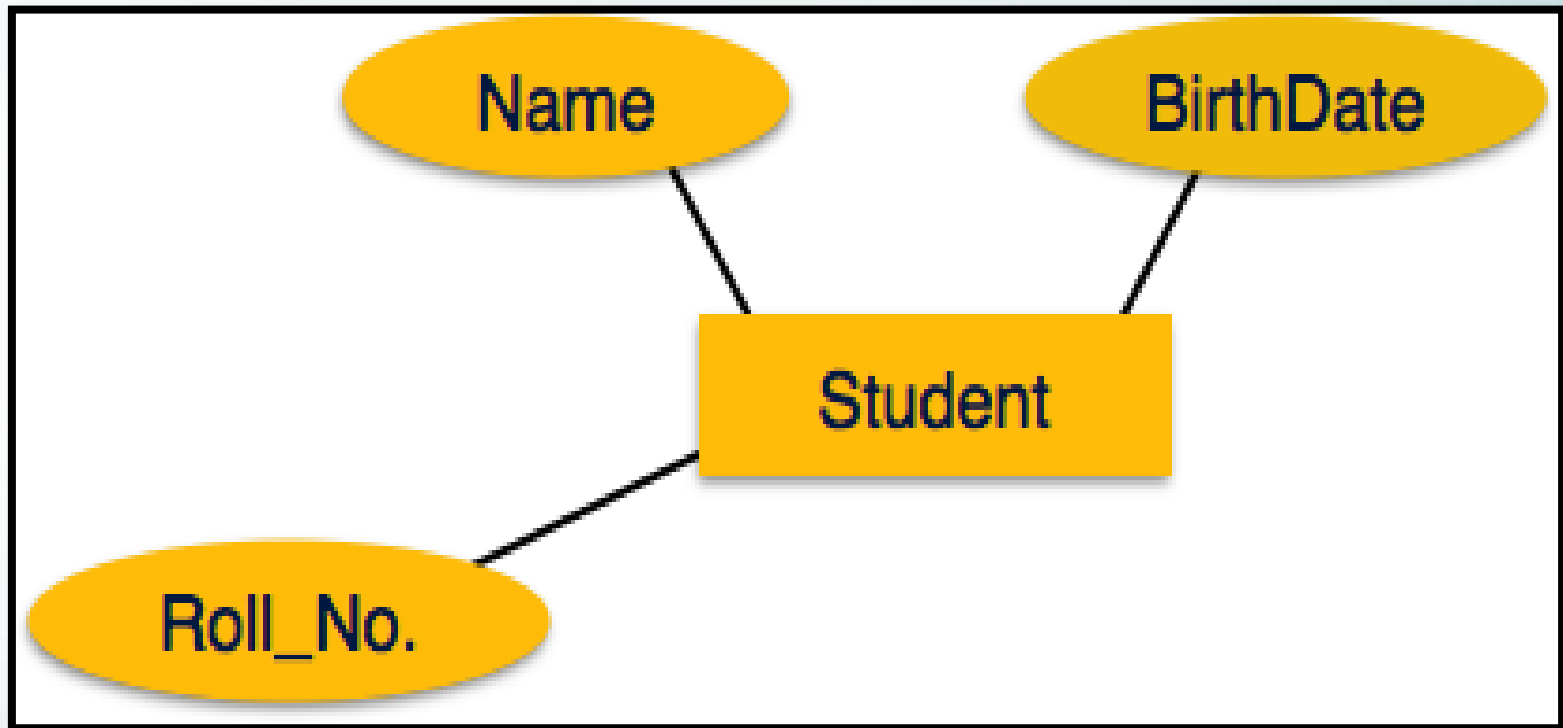
## ➤ **Attribute**

- Property of an entity or a relationship type.
- **For example**, a student entity may have name, class, and age as attributes.

## ➤ **Attribute Domain**

- Set of allowable values for one or more attributes.

## EXAMPLE: For an ATTRIBUTE



# ATTRIBUTES TYPES

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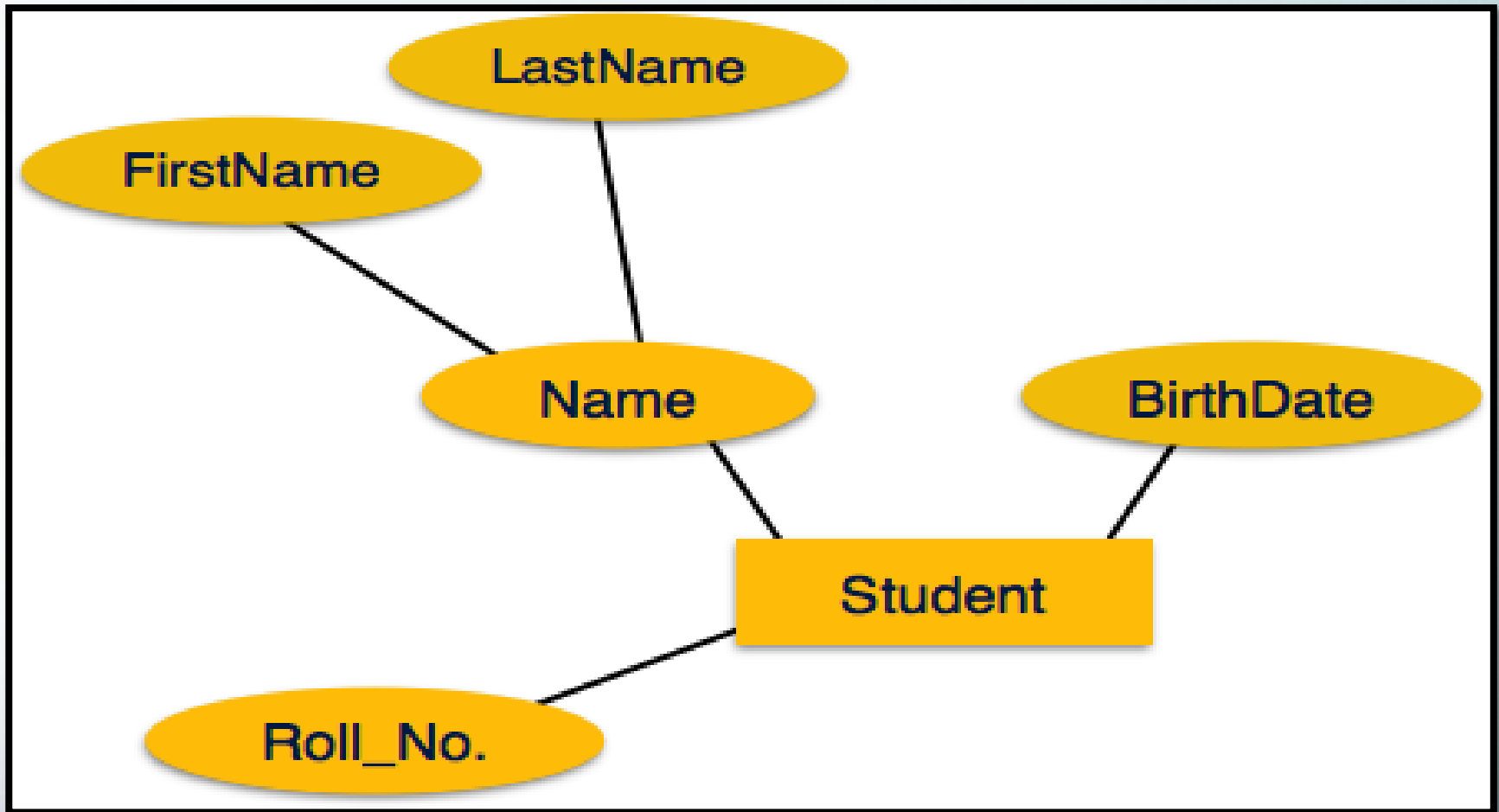
## ➤ **Simple Attribute**

- Attribute composed of a single component with an independent existence.
- **For example**, a student's phone number is an atomic value of 10 digits.

## ➤ **Composite Attribute**

- Attribute composed of multiple components, each with an independent existence.
- **For example**, a student's complete name may have first\_name and last\_name.

# EXAMPLE: COMPOSITE ATTRIBUTE



# ATTRIBUTES TYPES

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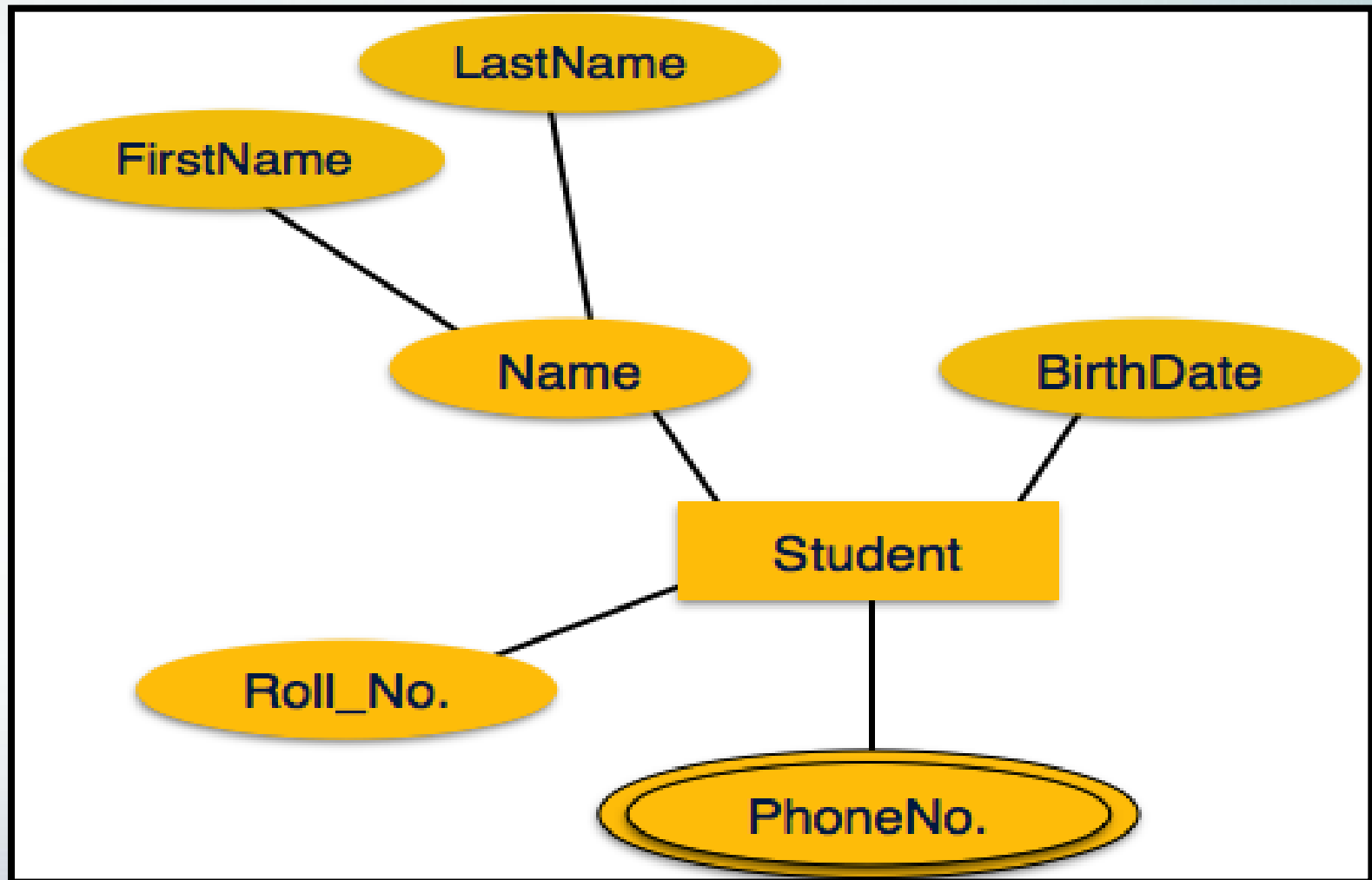
## ➤ **Single-valued Attribute**

- Attribute that holds a single value for each occurrence of an entity type.
- **For example** – Social\_Security\_Number.

## ➤ **Multi-valued Attribute**

- Attribute that holds multiple values for each occurrence of an entity type.
- **For example**, a person can have more than one phone number, email\_address, etc.

# EXAMPLE: MULTIVALUED ATTRIBUTE



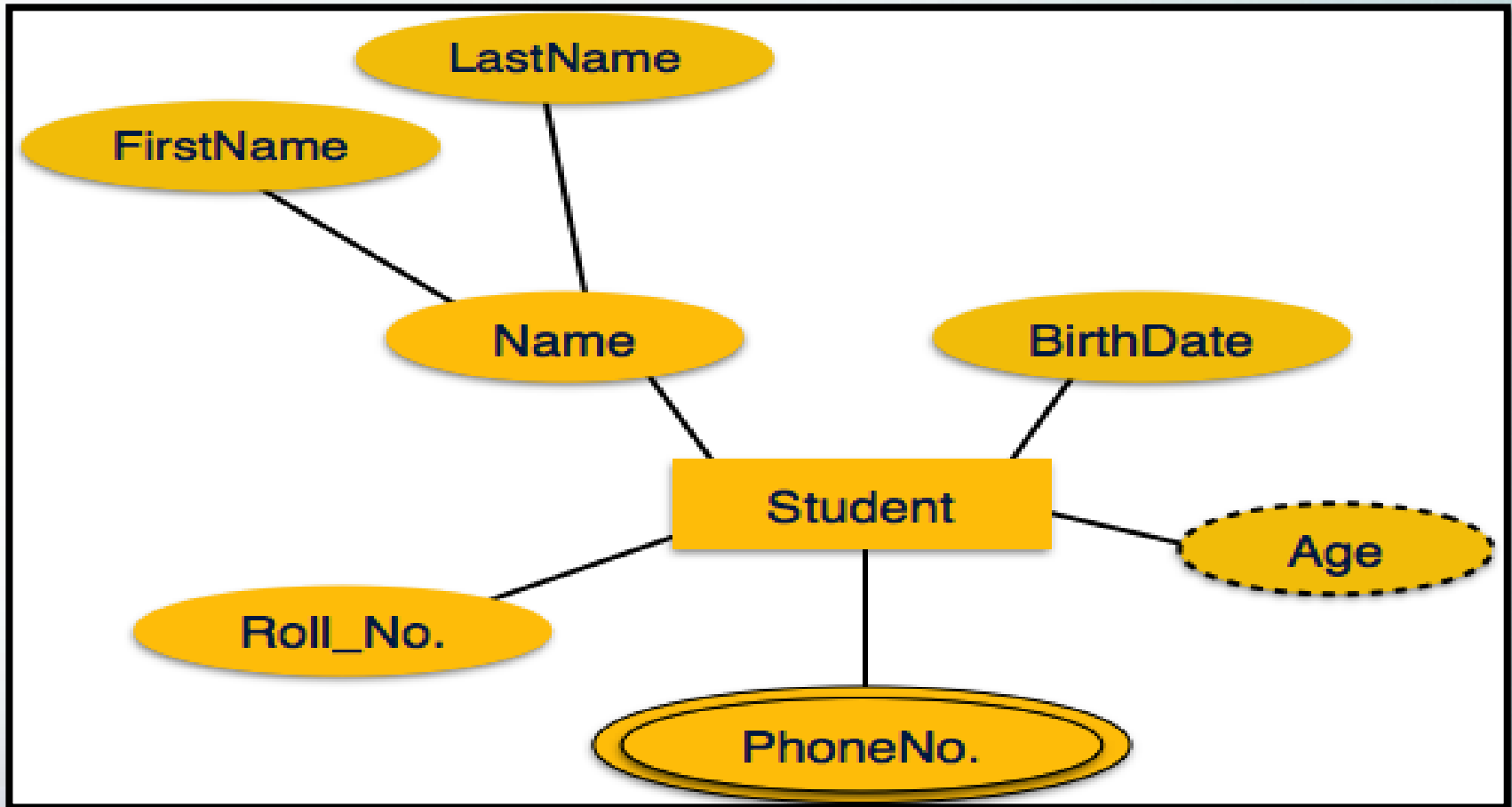
# ATTRIBUTES TYPES

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## ➤ **Derived Attribute**

- Attribute that represents a value that is derivable from value of a related attribute, or set of attributes, not necessarily in the same entity type.
- **For example,** age can be derived from data\_of\_birth.

# EXAMPLE: DERIVED ATTRIBUTE



# Entity-Set and Keys

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- **Key** is an attribute or collection of attributes that uniquely identifies an entity among entity set.
- **For example**, the roll number of a student makes him/her identifiable among students.
- **Super Key** – is defined as a set of attributes within a table that uniquely identifies each record within a table.
- **Candidate Key** – Candidate keys are defined as the set of fields from which primary key can be selected. It is an attribute or set of attribute that can act as a primary key for a table to uniquely identify each record in that table.

# Entity-Set and Keys

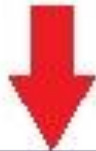
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- **Primary Key** – A primary key is one of the candidate keys chosen by the database designer to uniquely identify the entity set.
- **Composite Key** Key that consist of two or more attributes that uniquely identify an entity occurrence is called **Composite key**.

# EXAMPLE: PRIMARY KEY

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Primary Key



| s_id | S_name | age | course | address |
|------|--------|-----|--------|---------|
|      |        |     |        |         |

# EXAMPLE: COMPOSITE KEY

Composite Key



| cust_id | order_id | sale_detail |
|---------|----------|-------------|
|         |          |             |

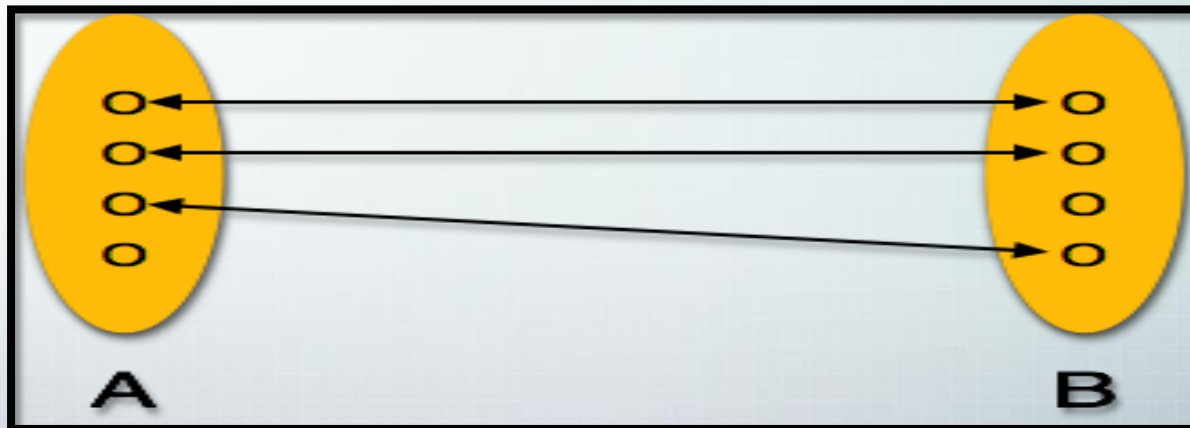
# Relationship

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- The association among entities is called a relationship. For example, an employee **works\_at** a department, a student **enrolls** in a course. Here, Works\_at and Enrolls are called relationships.
- **Relationship Set:**
  - A set of relationships of similar type is called a relationship set. Like entities, a relationship too can have attributes. *These attributes are called descriptive attributes.*

# Mapping Cardinalities

- **Cardinality** defines the number of entities in one entity set, which can be associated with the number of entities of other set via relationship set.
- **One-to-one** – One entity from entity set A can be associated with at most one entity of entity set B and vice versa.



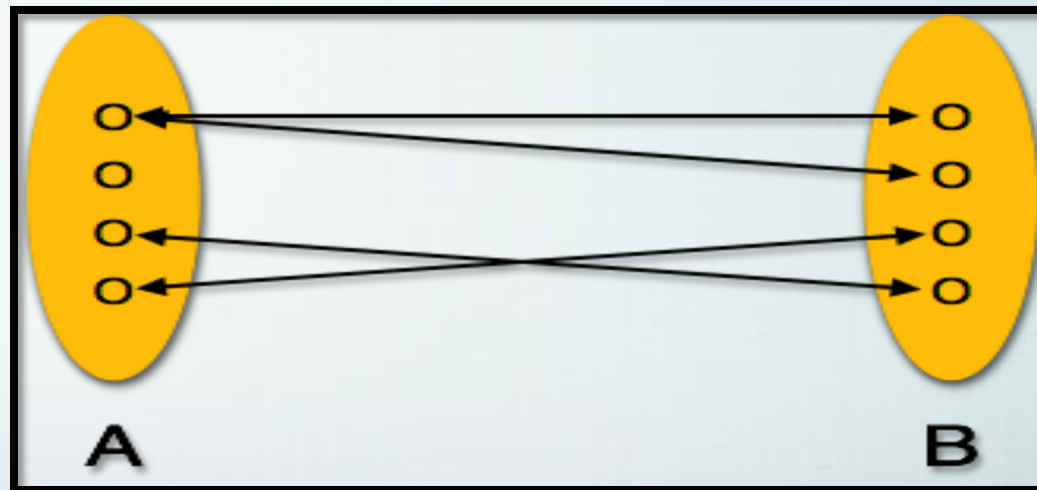
# EXAMPLE: 1 – 1

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# Mapping Cardinalities

- **One-to-many** – One entity from entity set A can be associated with more than one entities of entity set B however an entity from entity set B, can be associated with at most one entity.



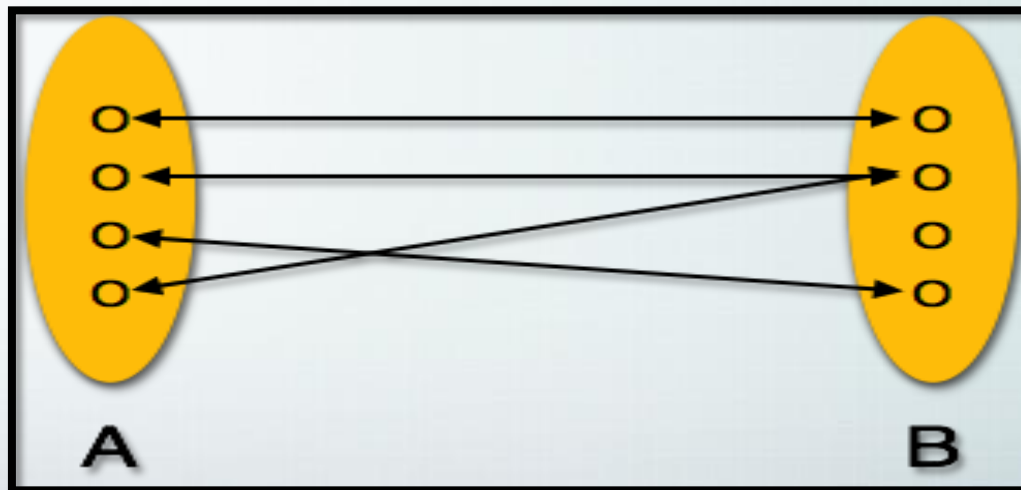
# EXAMPLE: 1 - N

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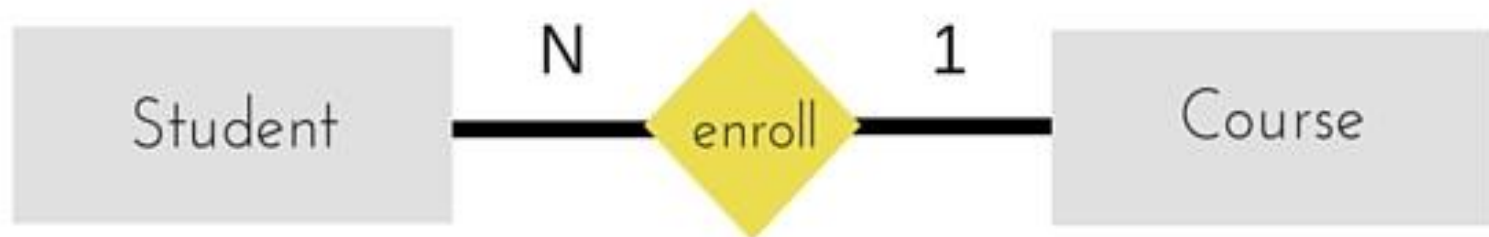


# Mapping Cardinalities

- **Many-to-one** – More than one entities from entity set A can be associated with at most one entity of entity set B, however an entity from entity set B can be associated with more than one entity from entity set A.

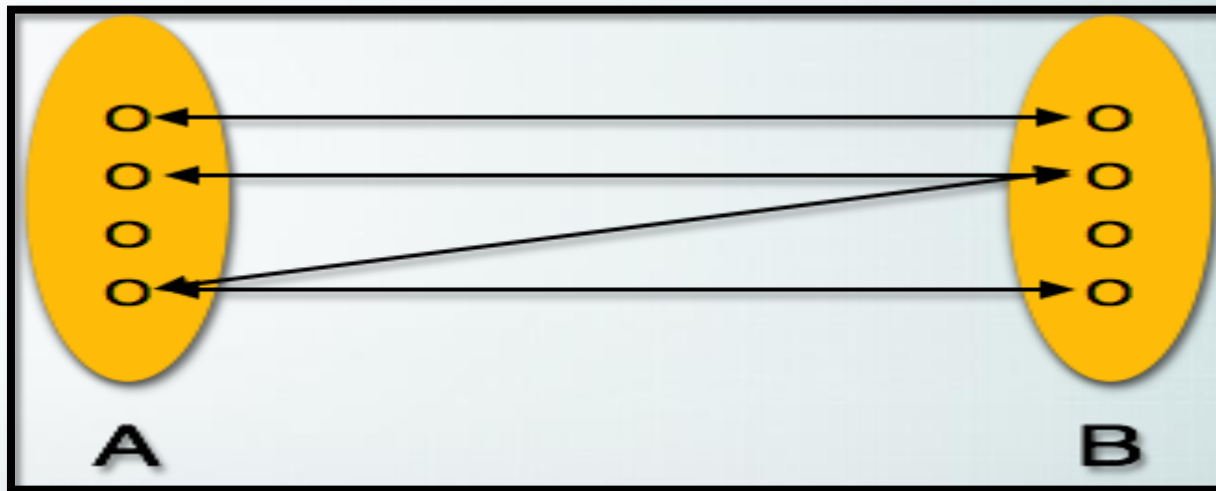


# EXAMPLE: N - 1

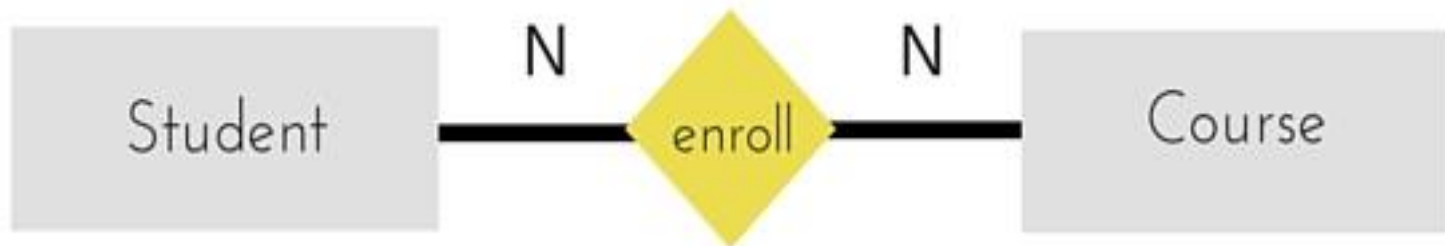


# Mapping Cardinalities

- **Many-to-many** – One entity from A can be associated with more than one entity from B and vice versa.



## EXAMPLE: N - N



# Normalisation

---

- Normalisation is a 'fancy' term for a set of rules, designed to make sure that a database is organised in the best way possible
- This allows the data to be processed more efficiently and any query to be processed.
- These rules depend on relationships being established between the entities to create a functional dependency between them.

# The normalisation process involves:

---

- **Finding and grouping together all the entities and their attributes.**
- **Removing repeating groups of data.**
- **Providing unique keys for each entity in the database system.**

# The Three Major Stages of Normalisation:

---

## ➤ **First Normal Form**

- 1NF is the first level of normalisation. An entity (table) is in First Normal form if it contains no repeating attributes (fields) or groups of attributes.

## ➤ **Second Normal Form**

- An entity is in 2NF if no attribute (not part of the primary key) is dependent on only part of the primary key. This only applies to entities with concatenated primary keys.

## ➤ **Third Normal Form**

- An entity is in 3NF if all attributes are entirely dependent on the primary key and not on any attribute that is not part of the primary key.
- In a relational schema, each tuple is divided into fields called Domains.

# Functional Dependency

---

**Functional dependency means that there must be only a one-to-one dependency for each attribute mapped from a primary key to that attribute.**

**It defines a relationship in which the existence of one entity/attribute is entirely dependent on the existence of another (one-to-one).**

# Functional Dependency - Practical Example

| SALES               |
|---------------------|
| <u>Order Number</u> |
| Acc.No.             |
| Customer            |
| Address             |
| Date                |
| Item                |
| Quantity            |
| Item Price          |
| Total Cost          |

Order Number is the primary key.

The value for each attribute of SALES, except Item Price, depends upon the value of the primary key.

All attributes of SALES, except Item Price, are Functionally Dependent on the primary key

Item Price is Functionally Dependent on the attribute Item

## To produce a set of entities in First Normal Form (1NF):

---

- Remove repeating (multiple) groups within the primary entities (tables) so that each record (row) within the entity is the same length.
- Repeating groups then become new entities, linked together by a one-to-many relationship.
- Relationships are created by including a primary key from one entity as a foreign key in another entity

# Al's Baker Shop

| Order No. | Acc. No. | Customer       | Address                 | Date | Item          | Qty. | Item Price | Total Cost |
|-----------|----------|----------------|-------------------------|------|---------------|------|------------|------------|
| 7823      | 178      | Daisy's Café   | 27 Bay Drive, Cove      | 16/7 | Bakewell Tart | 20   | 0.15       | 12.35      |
|           |          |                |                         |      | Danish Pastry | 13   | 0.20       |            |
|           |          |                |                         |      | Apple Pie     | 45   | 0.15       |            |
| 4633      | 526      | Smiths         | 12 Dee View, Aberdeen   | 19/7 | Butteries     | 120  | 0.20       | 24.00      |
| 2276      | 167      | Sally's Snacks | 3 High Street, Banchory | 17/7 | Apple Pie     | 130  | 0.15       | 56.50      |
|           |          |                |                         |      | Cherry Pie    | 100  | 0.18       |            |
|           |          |                |                         |      | Steak Pie     | 30   | 0.50       |            |
|           |          |                |                         |      | Meringue Pie  | 20   | 0.20       |            |
| 1788      | 032      | Tasty Bite     | 17 Wood Place, Inch     | 18/7 | Apple Pie     | 15   | 0.15       | 7.50       |
|           |          |                |                         |      | Danish Pastry | 50   | 0.20       |            |

# Al's Baker Shop

| Order No. | Acc. No. | Customer       | Address                 | Date | Item          | Qty. | Item Price | Total Cost |
|-----------|----------|----------------|-------------------------|------|---------------|------|------------|------------|
| 7823      | 178      | Daisy's Café   | 27 Bay Drive, Cove      | 16/7 | Bakewell Tart | 20   | 0.15       | 12.35      |
|           |          |                |                         |      | Danish Pastry | 13   | 0.20       |            |
|           |          |                |                         |      | Apple Pie     | 45   | 0.15       |            |
| 4633      | 526      | Smiths         | 12 Dee View, Aberdeen   | 19/7 | Butteries     | 120  | 0.20       | 24.00      |
| 2276      | 167      | Sally's Snacks | 3 High Street, Banchory | 17/7 | Apple Pie     | 130  | 0.15       | 56.50      |
|           |          |                |                         |      | Cherry Pie    | 100  | 0.18       |            |
|           |          |                |                         |      | Steak Pie     | 30   | 0.50       |            |
|           |          |                |                         |      | Meringue Pie  | 20   | 0.20       |            |
| 1788      | 032      | Tasty Bite     | 17 Wood Place, Inch     | 18/7 | Apple Pie     | 15   | 0.15       | 7.50       |
|           |          |                |                         |      | Danish Pastry | 50   | 0.20       |            |

# Al's Baker Shop

## Orders

| Order No. | Acc. No. | Customer       | Address                 | Date | Total Cost |
|-----------|----------|----------------|-------------------------|------|------------|
| 7823      | 178      | Daisy's Café   | 27 Bay Drive, Cove      | 16/7 | 12.35      |
| 4633      | 526      | Smiths         | 12 Dee View, Aberdeen   | 19/7 | 24.00      |
| 2276      | 167      | Sally's Snacks | 3 High Street, Banchory | 17/7 | 56.50      |
| 1788      | 032      | Tasty Bite     | 17 Wood Place, Inch     | 18/7 | 7.50       |

| Item          | Qty. | Item Price |
|---------------|------|------------|
| Bakewell Tart | 20   | 0.15       |
| Danish Pastry | 13   | 0.20       |
| Apple Pie     | 45   | 0.15       |
| Butteries     | 120  | 0.20       |
| Apple Pie     | 130  | 0.15       |
| Cherry Pie    | 100  | 0.18       |
| Steak Pie     | 30   | 0.50       |
| Meringue Pie  | 20   | 0.20       |
| Apple Pie     | 15   | 0.15       |
| Danish Pastry | 50   | 0.20       |

## Items Purchased

# Al's Baker Shop



| Order No. | Acc. No. | Customer       | Address                 | Date | Total Cost |
|-----------|----------|----------------|-------------------------|------|------------|
| 7823      | 178      | Daisy's Café   | 27 Bay Drive, Cove      | 16/7 | 12.35      |
| 4633      | 526      | Smiths         | 12 Dee View, Aberdeen   | 19/7 | 24.00      |
| 2276      | 167      | Sally's Snacks | 3 High Street, Banchory | 17/7 | 56.50      |
| 1788      | 032      | Tasty Bite     | 17 Wood Place, Inch     | 18/7 | 7.50       |

Orders

| Order No. | Item          | Qty. | Item Price |
|-----------|---------------|------|------------|
| 7823      | Bakewell Tart | 20   | 0.15       |
| 7823      | Danish Pastry | 13   | 0.20       |
| 7823      | Apple Pie     | 45   | 0.15       |
| 4633      | Butteries     | 120  | 0.20       |
| 2276      | Apple Pie     | 130  | 0.15       |
| 2276      | Cherry Pie    | 100  | 0.18       |
| 2276      | Steak Pie     | 30   | 0.50       |
| 2276      | Meringue Pie  | 20   | 0.20       |
| 1788      | Apple Pie     | 15   | 0.15       |
| 1788      | Danish Pastry | 50   | 0.20       |

Items Purchased

# Al's Baker Shop

Orders Table:

| Order No. | Acc. No. | Customer       | Address                 | Date | Total Cost |
|-----------|----------|----------------|-------------------------|------|------------|
| 7823      | 178      | Daisy's Café   | 27 Bay Drive, Cove      | 16/7 | 12.35      |
| 4633      | 526      | Smiths         | 12 Dee View, Aberdeen   | 19/7 | 24.00      |
| 2276      | 167      | Sally's Snacks | 3 High Street, Banchory | 17/7 | 56.50      |
| 1788      | 032      | Tasty Bite     | 17 Wood Place, Inch     | 18/7 | 7.50       |

Order No. can be used to uniquely identify each record and can therefore be made the primary key.

Orders

(Order No.  
Acc. No.  
Customer  
Address  
Date  
Total Cost)

# Al's Baker Shop

| Order No. | Item          | Qty. | Item Price |
|-----------|---------------|------|------------|
| 7823      | Bakewell Tart | 20   | 0.15       |
| 7823      | Danish Pastry | 13   | 0.20       |
| 7823      | Apple Pie     | 45   | 0.15       |
| 4633      | Butteries     | 120  | 0.20       |
| 2276      | Apple Pie     | 130  | 0.15       |
| 2276      | Cherry Pie    | 100  | 0.18       |
| 2276      | Steak Pie     | 30   | 0.50       |
| 2276      | Meringue Pie  | 20   | 0.20       |
| 1788      | Apple Pie     | 15   | 0.15       |
| 1788      | Danish Pastry | 50   | 0.20       |

Items Purchased Table:

No one attribute can be used to uniquely identify a record.

Concatenated key is required

Order No. and Item together can uniquely identify a record.

Items  
Purchased

(\*Order No.  
Item  
Quantity  
Item Price)

# Al's Baker Shop

## First Normal Form

---

|        |                                                                              |
|--------|------------------------------------------------------------------------------|
| Orders | ( <u>Order No.</u><br>Acc. No.<br>Customer<br>Address<br>Date<br>Total Cost) |
|--------|------------------------------------------------------------------------------|

|                    |                                                               |
|--------------------|---------------------------------------------------------------|
| Items<br>Purchased | (* <u>Order No.</u><br><u>Item</u><br>Quantity<br>Item Price) |
|--------------------|---------------------------------------------------------------|

# Q & A

---

1. What is Normalisation?
2. What does Normalisation process involves?
3. What is Ist Normal Form?
4. In a relational schema, each tuple is divided into fields called Domains. T/F

# The Three Major Stages of Normalisation:

---

## ➤ **First Normal Form**

- 1NF is the first level of normalisation. An entity (table) is in First Normal form if it contains no repeating attributes (fields) or groups of attributes.

## ➤ **Second Normal Form**

- An entity is in 2NF if no attribute (not part of the primary key) is dependent on only part of the primary key. This only applies to entries with concatenated primary keys.

## ➤ **Third Normal Form**

- An entity is in 3NF if all attributes are entirely dependent on the primary key and not on any attribute that is not part of the primary key.

## To produce a set of entities in Second Normal Form (2NF):

---

- Test for dependency by testing each particular attribute in turn to check that it can be uniquely identified by making use of all the primary key. This test need not be completed unless you have at least one table which requires a concatenated Primary Key
- Remove all partially dependent attributes to a new entity.

N.B. – A concatenated key occurs when you need two fields together in order to uniquely identify a record

# Al's Baker Shop

## First Normal Form

---

|        |                                                                              |
|--------|------------------------------------------------------------------------------|
| Orders | ( <u>Order No.</u><br>Acc. No.<br>Customer<br>Address<br>Date<br>Total Cost) |
|--------|------------------------------------------------------------------------------|

|                    |                                                               |
|--------------------|---------------------------------------------------------------|
| Items<br>Purchased | (* <u>Order No.</u><br><u>Item</u><br>Quantity<br>Item Price) |
|--------------------|---------------------------------------------------------------|

# Al's Baker Shop

Orders Table:

| Order No. | Acc. No. | Customer       | Address                 | Date | Total Cost |
|-----------|----------|----------------|-------------------------|------|------------|
| 7823      | 178      | Daisy's Café   | 27 Bay Drive, Cove      | 16/7 | 12.35      |
| 4633      | 526      | Smiths         | 12 Dee View, Aberdeen   | 19/7 | 24.00      |
| 2276      | 167      | Sally's Snacks | 3 High Street, Banchory | 17/7 | 56.50      |
| 1788      | 032      | Tasty Bite     | 17 Wood Place, Inch     | 18/7 | 7.50       |

Because this entity has a single attribute as the primary key there can be no partial dependencies and therefore the entity is already in 2NF.

# Al's Baker Shop

## (2NF Step 1)

| Order No. | Item          | Qty. | Item Price |
|-----------|---------------|------|------------|
| 7823      | Bakewell Tart | 20   | 0.15       |
| 7823      | Danish Pastry | 13   | 0.20       |
| 7823      | Apple Pie     | 45   | 0.15       |
| 4633      | Butteries     | 120  | 0.20       |
| 2276      | Apple Pie     | 130  | 0.15       |
| 2276      | Cherry Pie    | 100  | 0.18       |
| 2276      | Steak Pie     | 30   | 0.50       |
| 2276      | Meringue Pie  | 20   | 0.20       |
| 1788      | Apple Pie     | 15   | 0.15       |
| 1788      | Danish Pastry | 50   | 0.20       |

Items Purchased Table:  
Primary key is Order No. and Item

Test for dependency by testing each particular attribute.

| Primary Key | Attribute | Functionally Dependent? |
|-------------|-----------|-------------------------|
|             |           |                         |
|             |           |                         |

# Al's Baker Shop

## (2NF Step 1)

| Order No. | Item          | Qty. | Item Price |
|-----------|---------------|------|------------|
| 7823      | Bakewell Tart | 20   | 0.15       |
| 7823      | Danish Pastry | 13   | 0.20       |
| 7823      | Apple Pie     | 45   | 0.15       |
| 4633      | Butteries     | 120  | 0.20       |
| 2276      | Apple Pie     | 130  | 0.15       |
| 2276      | Cherry Pie    | 100  | 0.18       |
| 2276      | Steak Pie     | 30   | 0.50       |
| 2276      | Meringue Pie  | 20   | 0.20       |
| 1788      | Apple Pie     | 15   | 0.15       |
| 1788      | Danish Pastry | 50   | 0.20       |

Items Purchased Table:  
Primary key is Order No. and Item

Test for dependency by testing each particular attribute.

| Primary Key      | Attribute | Functionally Dependent?                                          |
|------------------|-----------|------------------------------------------------------------------|
| Order No<br>Item | Quantity  | YES<br>Quantity is functionally dependent on Order No. and Item. |
|                  |           |                                                                  |

# AI's Baker Shop

## (2NF Step 1)

| Order No. | Item          | Qty. | Item Price |
|-----------|---------------|------|------------|
| 7823      | Bakewell Tart | 20   | 0.15       |
| 7823      | Danish Pastry | 13   | 0.20       |
| 7823      | Apple Pie     | 45   | 0.15       |
| 4633      | Butteries     | 120  | 0.20       |
| 2276      | Apple Pie     | 130  | 0.15       |
| 2276      | Cherry Pie    | 100  | 0.18       |
| 2276      | Steak Pie     | 30   | 0.50       |
| 2276      | Meringue Pie  | 20   | 0.20       |
| 1788      | Apple Pie     | 15   | 0.15       |
| 1788      | Danish Pastry | 50   | 0.20       |

Items Purchased Table:  
Primary key is Order No. and Item

Test for dependency by testing each particular attribute.

| Primary Key                   | Attribute  | Functionally Dependent?                                                           |
|-------------------------------|------------|-----------------------------------------------------------------------------------|
| Order No<br>Item<br>dependent | Quantity   | YES<br>Quantity is functionally<br>on Order No. and Item.                         |
| Order No<br>Item              | Item Price | NO<br>Item price is functionally dependent<br>Item, but not on Order No. and Item |

# Al's Baker Shop

## (2NF Step 2)

Remove any partially dependent attributes to a new entity

| Order No. | Item          | Qty. | Item Price |
|-----------|---------------|------|------------|
| 7823      | Bakewell Tart | 20   | 0.15       |
| 7823      | Danish Pastry | 13   | 0.20       |
| 7823      | Apple Pie     | 45   | 0.15       |
| 4633      | Butteries     | 120  | 0.20       |
| 2276      | Apple Pie     | 130  | 0.15       |
| 2276      | Cherry Pie    | 100  | 0.18       |
| 2276      | Steak Pie     | 30   | 0.50       |
| 2276      | Meringue Pie  | 20   | 0.20       |
| 1788      | Apple Pie     | 15   | 0.15       |
| 1788      | Danish Pastry | 50   | 0.20       |

| Item Price |
|------------|
| 0.15       |
| 0.20       |
| 0.15       |
| 0.20       |
| 0.15       |
| 0.18       |
| 0.50       |
| 0.20       |
| 0.15       |
| 0.20       |

# Al's Baker Shop

## (2NF Step 2)

Remove any partially dependent attributes to a new entity

| Order No. | Item          | Qty. |
|-----------|---------------|------|
| 7823      | Bakewell Tart | 20   |
| 7823      | Danish Pastry | 13   |
| 7823      | Apple Pie     | 45   |
| 4633      | Butteries     | 120  |
| 2276      | Apple Pie     | 130  |
| 2276      | Cherry Pie    | 100  |
| 2276      | Steak Pie     | 30   |
| 2276      | Meringue Pie  | 20   |
| 1788      | Apple Pie     | 15   |
| 1788      | Danish Pastry | 50   |

Part Order

| Item Price |
|------------|
| 0.15       |
| 0.15       |
| 0.20       |
| 0.18       |
| 0.20       |
| 0.20       |
| 0.50       |

Price List

# Al's Baker Shop

## (2NF Step 2)

Create a relationship between the tables  
and assign Primary Keys

| Order No. | Item          | Qty. |
|-----------|---------------|------|
| 7823      | Bakewell Tart | 20   |
| 7823      | Danish Pastry | 13   |
| 7823      | Apple Pie     | 45   |
| 4633      | Butteries     | 120  |
| 2276      | Apple Pie     | 130  |
| 2276      | Cherry Pie    | 100  |
| 2276      | Steak Pie     | 30   |
| 2276      | Meringue Pie  | 20   |
| 1788      | Apple Pie     | 15   |
| 1788      | Danish Pastry | 50   |

| Item          | Item Price |
|---------------|------------|
| Apple Pie     | 0.15       |
| Bakewell Tart | 0.15       |
| Butteries     | 0.20       |
| Cherry Pie    | 0.18       |
| Danish Pastry | 0.20       |
| Meringue Pie  | 0.20       |
| Steak Pie     | 0.50       |

Part Order  
Primary Key: Order No.  
and \*Item

Price List  
Primary Key: Item

# Al's Baker Shop

## Second Normal Form

---

|        |                                                                              |
|--------|------------------------------------------------------------------------------|
| Orders | ( <u>Order No.</u><br>Acc. No.<br>Customer<br>Address<br>Date<br>Total Cost) |
|--------|------------------------------------------------------------------------------|

|            |                              |
|------------|------------------------------|
| Price List | ( <u>Item</u><br>Item Price) |
|------------|------------------------------|

|            |                                                   |
|------------|---------------------------------------------------|
| Part Order | (* <u>Order No.</u><br>* <u>Item</u><br>Quantity) |
|------------|---------------------------------------------------|

# The Three Major Stages of Normalisation:

---

## ❖ First Normal Form

- 1NF is the first level of normalisation. An entity (table) is in First Normal form if it contains no repeating attributes (fields) or groups of attributes.

## ❖ Second Normal Form

- An entity is in 2NF if no attribute (not part of the primary key) is dependent on only part of the primary key. This only applies to entries with concatenated primary keys.

## ❖ Third Normal Form

- An entity is in 3NF if all attributes are entirely dependent on the primary key and not on any attribute that is not part of the primary key.

## To produce a set of entities in Third Normal Form (3NF):

---

- **Test each attribute in turn to check for dependency on the primary key.**
- **Remove all transitive dependencies to a new entity.**
  - A transitive dependency is where an attribute is dependent on another attribute (or attributes) that is (are) NOT the primary key

# Al's Baker Shop

## Second Normal Form

---

|        |                                                                              |
|--------|------------------------------------------------------------------------------|
| Orders | ( <u>Order No.</u><br>Acc. No.<br>Customer<br>Address<br>Date<br>Total Cost) |
|--------|------------------------------------------------------------------------------|

|            |                              |
|------------|------------------------------|
| Price List | ( <u>Item</u><br>Item Price) |
|------------|------------------------------|

|            |                                                   |
|------------|---------------------------------------------------|
| Part Order | (* <u>Order No.</u><br>* <u>Item</u><br>Quantity) |
|------------|---------------------------------------------------|

# Al's Baker Shop - 3NF Step1

Test for  
dependency

| Order No. | Acc. No. | Customer       | Address                 | Date | Total Cost |
|-----------|----------|----------------|-------------------------|------|------------|
| 7823      | 178      | Daisy's Café   | 27 Bay Drive, Cove      | 16/7 | 12.35      |
| 4633      | 526      | Smiths         | 12 Dee View, Aberdeen   | 19/7 | 24.00      |
| 2276      | 167      | Sally's Snacks | 3 High Street, Banchory | 17/7 | 56.50      |
| 1788      | 032      | Tasty Bite     | 17 Wood Place, Inch     | 18/7 | 7.50       |

| Primary Key | Attribute | Transitive Dependency |
|-------------|-----------|-----------------------|
|             |           |                       |

# Al's Baker Shop - 3NF Step1

Test for  
dependency

| Order No. | Acc. No. | Customer       | Address                 | Date | Total Cost |
|-----------|----------|----------------|-------------------------|------|------------|
| 7823      | 178      | Daisy's Café   | 27 Bay Drive, Cove      | 16/7 | 12.35      |
| 4633      | 526      | Smiths         | 12 Dee View, Aberdeen   | 19/7 | 24.00      |
| 2276      | 167      | Sally's Snacks | 3 High Street, Banchory | 17/7 | 56.50      |
| 1788      | 032      | Tasty Bite     | 17 Wood Place, Inch     | 18/7 | 7.50       |

| Primary Key | Attribute | Trasnsitive Dependency                                         |
|-------------|-----------|----------------------------------------------------------------|
| Order No.   | Acc.No.   | YES: Acc.No can be found if we know either Customer or Address |

## Al's Baker Shop - 3NF Step1

Test for  
dependency

| Order No. | Acc. No. | Customer       | Address                 | Date | Total Cost |
|-----------|----------|----------------|-------------------------|------|------------|
| 7823      | 178      | Daisy's Café   | 27 Bay Drive, Cove      | 16/7 | 12.35      |
| 4633      | 526      | Smiths         | 12 Dee View, Aberdeen   | 16/7 | 24.00      |
| 2276      | 167      | Sally's Snacks | 3 High Street, Banchory | 17/7 | 56.50      |
| 1788      | 032      | Tasty Bite     | 17 Wood Place, Inch     | 18/7 | 7.50       |

| Primary Key | Attribute | Transitive Dependency                                           |
|-------------|-----------|-----------------------------------------------------------------|
| Order No.   | Acc.No.   | YES: Customer can be found if we know either Acc.No. or Address |
| Order No.   | Customer  | YES Customer can be found if we know either Acc.No. or Address  |

## Al's Baker Shop - 3NF Step1

Test for  
dependency

| Order No. | Acc. No. | Customer       | Address                 | Date | Total Cost |
|-----------|----------|----------------|-------------------------|------|------------|
| 7823      | 178      | Daisy's Café   | 27 Bay Drive, Cove      | 16/7 | 12.35      |
| 4633      | 526      | Smiths         | 12 Dee View, Aberdeen   | 16/7 | 24.00      |
| 2276      | 167      | Sally's Snacks | 3 High Street, Banchory | 17/7 | 56.50      |
| 1788      | 032      | Tasty Bite     | 17 Wood Place, Inch     | 18/7 | 7.50       |

| Primary Key | Attribute | Transitive Dependency                                           |
|-------------|-----------|-----------------------------------------------------------------|
| Order No.   | Acc.No.   | YES: Customer can be found if we know either Acc.No. or Address |
| Order No.   | Customer  | YES Acc.No. can be found if we know either Customer or Address  |
| Order No.   | Address   | YES Address can be found if we know either Customer or Acc.No   |

## Al's Baker Shop - 3NF Step1

Test for  
dependency

| Order No. | Acc. No. | Customer       | Address                 | Date | Total Cost |
|-----------|----------|----------------|-------------------------|------|------------|
| 7823      | 178      | Daisy's Café   | 27 Bay Drive, Cove      | 16/7 | 12.35      |
| 4633      | 526      | Smiths         | 12 Dee View, Aberdeen   | 19/7 | 24.00      |
| 2276      | 167      | Sally's Snacks | 3 High Street, Banchory | 17/7 | 56.50      |
| 1788      | 032      | Tasty Bite     | 17 Wood Place, Inch     | 18/7 | 7.50       |

| Primary Key | Attribute | Transitive Dependency                                           |
|-------------|-----------|-----------------------------------------------------------------|
| Order No.   | Acc.No.   | YES: Customer can be found if we know either Acc.No. or Address |
| Order No.   | Customer  | YES Acc.No. can be found if we know either Customer or Address  |
| Order No.   | Address   | YES Address can be found if we know either Customer or Acc.No   |
| Order No.   | Date      | NO Dependent on Order No.                                       |

## Al's Baker Shop - 3NF Step1

Test for  
dependency

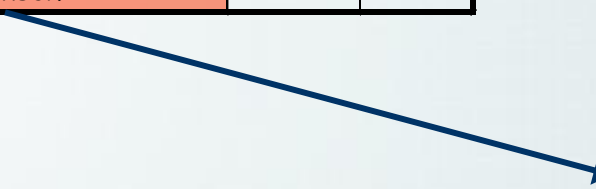
| Order No. | Acc. No. | Customer       | Address                 | Date | Total Cost |
|-----------|----------|----------------|-------------------------|------|------------|
| 7823      | 178      | Daisy's Café   | 27 Bay Drive, Cove      | 16/7 | 12.35      |
| 4633      | 526      | Smiths         | 12 Dee View, Aberdeen   | 19/7 | 24.00      |
| 2276      | 167      | Sally's Snacks | 3 High Street, Banchory | 17/7 | 56.50      |
| 1788      | 032      | Tasty Bite     | 17 Wood Place, Inch     | 18/7 | 7.50       |

| Primary Key | Attribute  | Transitive Dependency                                           |
|-------------|------------|-----------------------------------------------------------------|
| Order No.   | Acc.No.    | YES: Customer can be found if we know either Acc.No. or Address |
| Order No.   | Customer   | YES Acc.No. can be found if we know either Customer or Address  |
| Order No.   | Address    | YES Address can be found if we know either Customer or Acc.No   |
| Order No.   | Date       | NO Dependent on Order No.                                       |
| Order No.   | Total Cost | NO Dependent on Order No.                                       |

# Al's Baker Shop - 3NF Step2

Remove transitive dependencies to a new entity

| Order No. | Acc. No. | Customer       | Address                 | Date | Total Cost |
|-----------|----------|----------------|-------------------------|------|------------|
| 7823      | 178      | Daisy's Café   | 27 Bay Drive, Cove      | 16/7 | 12.35      |
| 4633      | 526      | Smiths         | 12 Dee View, Aberdeen   | 16/7 | 24.00      |
| 2276      | 167      | Sally's Snacks | 3 High Street, Banchory | 17/7 | 56.50      |
| 1788      | 032      | Tasty Bite     | 17 Wood Place, Inch     | 18/7 | 7.50       |



| Acc. No. | Customer       | Address                 |
|----------|----------------|-------------------------|
| 178      | Daisy's Café   | 27 Bay Drive, Cove      |
| 526      | Smiths         | 12 Dee View, Aberdeen   |
| 167      | Sally's Snacks | 3 High Street, Banchory |
| 032      | Tasty Bite     | 17 Wood Place, Inch     |

# Al's Baker Shop - 3NF Step2

Remove transitive dependencies to a new entity

## Orders

| Order No. | Date | Total Cost |
|-----------|------|------------|
| 7823      | 16/7 | 12.35      |
| 4633      | 16/7 | 24.00      |
| 2276      | 17/7 | 56.50      |
| 1788      | 18/7 | 7.50       |

## Customers

| Acc. No. | Customer       | Address                 |
|----------|----------------|-------------------------|
| 178      | Daisy's Café   | 27 Bay Drive, Cove      |
| 526      | Smiths         | 12 Dee View, Aberdeen   |
| 167      | Sally's Snacks | 3 High Street, Banchory |
| 032      | Tasty Bite     | 17 Wood Place, Inch     |

# Al's Baker Shop - 3NF Step2

Create a relationship between the tables  
and assign Primary Keys

Orders

| Order No. | Acc. No. | Date | Total Cost |
|-----------|----------|------|------------|
| 7823      | 178      | 16/7 |            |
| 4633      | 526      | 16/7 |            |
| 2276      | 167      | 17/7 |            |
| 1788      | 032      | 18/7 |            |

Primary Key:  
Order No.

Customers

| Acc. No. | Customer       | Address                 |
|----------|----------------|-------------------------|
| 178      | Daisy's Café   | 27 Bay Drive, Cove      |
| 526      | Smiths         | 12 Dee View, Aberdeen   |
| 167      | Sally's Snacks | 3 High Street, Banchory |
| 032      | Tasty Bite     | 17 Wood Place, Inch     |

Primary Key:  
Acc.No.

# Al's Baker Shop

## Third Normal Form

|           |                                           |
|-----------|-------------------------------------------|
| Customers | ( <u>Acc. No.</u><br>Customer<br>Address) |
|-----------|-------------------------------------------|

|        |                                                        |
|--------|--------------------------------------------------------|
| Orders | ( <u>Order No.</u><br>*Acc. No.<br>Date<br>Total Cost) |
|--------|--------------------------------------------------------|

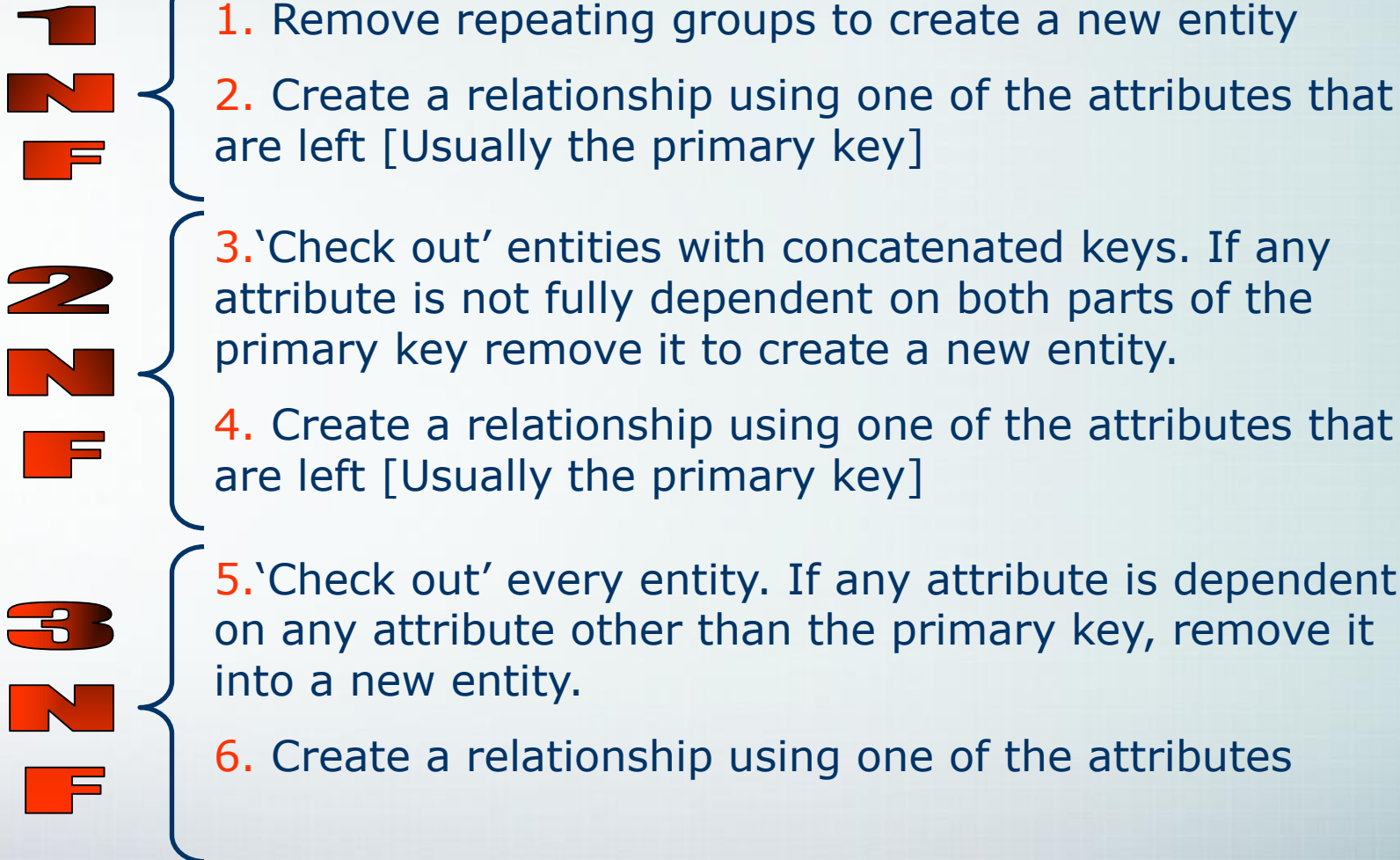
|            |                              |
|------------|------------------------------|
| Price List | ( <u>Item</u><br>Item Price) |
|------------|------------------------------|

|            |                                                   |
|------------|---------------------------------------------------|
| Part Order | (* <u>Order No.</u><br>* <u>Item</u><br>Quantity) |
|------------|---------------------------------------------------|

## Normalisation Complete

# The Six Point Plan

## Normalisation -

- 
- The diagram illustrates the Six Point Plan for Normalisation. It features a vertical list of six points on the right, grouped by three large curly braces on the left. The first brace groups points 1 and 2, the second brace groups points 3 and 4, and the third brace groups points 5 and 6. To the left of each brace is a large, stylized number: '1' for the first group, '2' for the second, and '3' for the third. Each of these large numbers is composed of a smaller '1', a '2', and an 'F' stacked vertically.
1. Remove repeating groups to create a new entity
  2. Create a relationship using one of the attributes that are left [Usually the primary key]
  3. 'Check out' entities with concatenated keys. If any attribute is not fully dependent on both parts of the primary key remove it to create a new entity.
  4. Create a relationship using one of the attributes that are left [Usually the primary key]
  5. 'Check out' every entity. If any attribute is dependent on any attribute other than the primary key, remove it into a new entity.
  6. Create a relationship using one of the attributes

## Q & A

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1. What is 2<sup>nd</sup> Normal Form?
2. In 2<sup>nd</sup> Normal Form which dependency is tested?
3. What is 3<sup>rd</sup> Normal Form?
4. What is Transitive Dependency?